SECOND REVIEW · ACADEMIC PROTOTYPE · NOT AN OPERATIONAL SYSTEM

Component D — Topics and speed

How posts in a window become themes, and which themes are speeding up.

This brief is for the teammate who presents topics. A theme is a cluster of similar posts. Speed is posts per hour, using the time the message was posted.

1. What this agent does

Component D groups recent posts into themes and ranks those themes by how fast they are being posted, and by whether that speed is rising or falling. It does not decide who is coordinated. That is Component E.

2. Where it sits

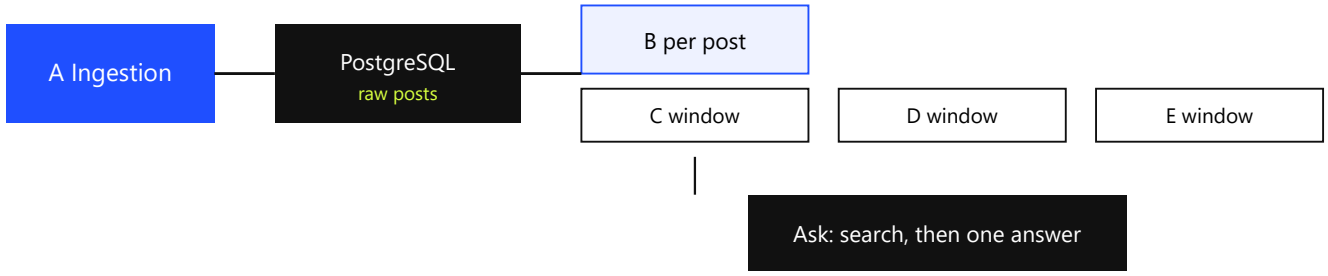


Figure 1. Posts land in Postgres. B reads one post at a time. C, D, and E recompute a time window. Ask searches the short summaries those agents write.

D recomputes the whole window each time. At this project's size, a few hundred to a few thousand posts, a full recompute is simpler than an incremental topic model and is what the brief locks.

3. How a theme is made

1. Take posts in the window that have text. The clock is `posted_at`.
2. Embed each post with `paraphrase-multilingual-MiniLM-L12-v2`. The same model embeds Ask queries. Dimension 384.
3. Cluster with BERTopic and HDBSCAN. Minimum topic size is 3, so a tiny corpus can still form a theme.
4. Cap the run at 500 documents so memory stays bounded.
5. Name the theme from its distinctive words.

4. Speed and change

Split the window in half. Velocity is posts per hour across the whole window. Acceleration compares the second half with the first. Positive acceleration is the "rising" mark on the Trends page. Negative acceleration means the theme is cooling. Neither number is a prediction of tomorrow.

Symbol	Plain meaning
v	Posts in the theme, divided by hours in the window.

a	How much v changed from the first half of the window to the second.
Rising	a is above zero.



Figure 2. A theme is rising only if it is posted faster in the later half than in the earlier half.

5. What to show

Open **Trends**. Read one theme name, its posts per hour, and whether it is rising. Say the words in the theme are extracted terms, not a headline written by a person.

> **One sentence.** “We measure how fast a cluster of similar posts is arriving, on the clock Telegram stamped, not on the clock we downloaded them.”

6. Related parts

Part	Relation to D
Table	`trend_topics_d`
API	`GET /api/trends`. The page shows the latest run in the window, not every past run stacked together.
Ask	Can cite a topic summary. If nothing is close enough, Ask refuses instead of inventing a theme.
B	Not required. D clusters raw text. Sentiment is a different page.

7. Done, and not done

Done	Not claimed
BERTopic path, 500-document cap, velocity and acceleration, Trends page.	No manual theme-quality study yet. That is Phase 5. A theme can be messy when the window is small.

> **Do not say** the model “understood the news”. It grouped similar wording and counted speed.

8. Likely questions

Question	Answer
Why minimum size 3?	A student corpus is small. A higher minimum would hide every theme.
Why 500?	BERTopic embeddings spike memory. The cap is a hardware limit, stated in the brief.
Why did topics not change after a download?	D writes a snapshot when analysis runs. New posts sit in the database until the next window run.